



DATE

23 September 2026

SOLAR SYSTEM AUDIT REPORT

CUSTOMER

Mr Stephen Ukaegbu

SYSTEM

6 kW Hybrid Inverter / 10 kWh Lithium Battery

FAULT

Undersized Solar Panel Array

CONNECTED LOADS

- 5 Ceiling Fans
- 2 Freezers
- 1 HP Inverter Air Conditioner

OPERATING HOURS

The system is used mainly during the daytime and evening hours. The connected appliances are not operated simultaneously, and the actual load varies depending on the appliances in use.

OBSERVATION

During system inspection, it was observed that while the existing PV capacity is undersized to adequately support the 6 kW hybrid inverter and 10 kWh lithium battery system under certain operating conditions, it remains operational under lower or variable load conditions.

However, the existing PV capacity does not provide sufficient capacity to achieve the customer's desired level of battery charging and extended system uptime. A corrective replacement of the existing PV capacity will improve battery charging performance and further increase the uptime of the inverter and battery system for the customer.

RECOMMENDATION

The existing solar panel array should be corrected by increasing the total PV capacity to increase the uptime of the inverter and battery system. This will improve charging performance and further extend the available operating hours of the system during normal solar hours.

CONCLUSION

The current PV capacity needs to be increased to meet the desired capacity of the customer for increased uptime hours.